# Graduate Mathematics in the Age of AI

Forming Mathematicians for Original, Independent, and Responsible Inquiry

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**Abstract**

Artificial intelligence can increasingly produce plausible, technically sophisticated mathematical material faster than a developing graduate student can understand or verify it. A sophisticated result or paper draft consequently becomes weaker evidence of the student's own mathematical development. This creates a formation gap between output and personal capacity, and a trust gap between a convincing argument and a warranted basis for accepting it. The formation gap can persist even when the student genuinely understands the output: understanding a supplied argument does not by itself establish the capacity to initiate and direct inquiry.

These gaps are not the whole educational story. AI can also help students explore more examples, compare approaches, enter unfamiliar areas, and undertake ambitious research. The task is to design an apprenticeship that realizes these possibilities while developing substantive mathematical command. The central purpose of a mathematics PhD should be the formation of mathematicians capable of original, independent, and responsible inquiry, including inquiry conducted in collaboration with AI.

This document develops that objective through four connected capacities: competence, judgment, independence, and responsibility. It distinguishes a work's contribution to mathematics from the evidence it provides of a student's formation; explains how a known answer can initiate rather than end creative inquiry; and proposes changes in learning activities, assessment, doctoral originality, advising, and institutional support. Purposeful independent work and ambitious AI-assisted research are complementary parts of the model. Its practical recommendations include proportionate contribution statements, explicit recognition of advising costs, and staged pilots that evaluate both internal mathematical ability and effective human–AI collaboration. The aim is not to preserve an inherited sequence of training, but to improve mathematical formation as the means of doing mathematics change.

# Contents


**I The apprenticeship under pressure** **4**

**1 Purpose and premise** **4**

**2 The traditional apprenticeship** **4**

2.1 Formation before productivity . . . . . 5

2.2 From student to contributor . . . . . 5

**3 The formation gap** **5**

**4 The trust gap** **6**

**5 The challenges for students, advisors, and graduate programs** **8**

5.1 For the student . . . . . 8

5.2 For the advisor . . . . . 8

5.3 For the program . . . . . 9

**II Why human mathematical capacity still matters** **10**

**6 The case for continued formation** **10**

6.1 Discovery, judgment, and synthesis . . . . . 10

6.2 Institutional stewardship and intellectual agency . . . . . 11

6.3 Continuity, resilience, and option value . . . . . 11

**7 A new educational objective** **11**

**III A proposed approach to training and advising** **13**

**8 Designing learning with powerful tools** **13**

**9 Knowing an answer as a beginning** **13**

**10 Verification, independence, and community** **14**

**11 Roles and conditions of AI use** **14**

**12 An integrated training model** **16**

12.1 Knowledge, technique, and mathematical orientation . . . . . 16
12.2 Guided study and research apprenticeship . . . . . 16
12.3 Progression toward research leadership . . . . . 17

**13 The advisor in the new model** **17**

## IV Assessment and program redesign 19

**14 Assessing capacities rather than artifacts alone** **19**

**15 Exercises in judgment and creative inquiry** **20**

**16 Milestones, examinations, and the dissertation** **20**
16.1 Originality and the student's contribution . . . . . 21
16.2 Evidence, defense, and proportionate records . . . . . 21

**17 Curriculum, policy, and community** **22**

**18 Duration, scale, and professional preparation** **22**

## V Institutional responsibility and implementation 24

**19 Recognizing the work of advising** **24**

**20 A staged departmental implementation** **24**

**21 Learning from a pilot** **25**

**22 Conclusion** **27**

**AI assistance disclosure** **27**

**A A sample protected-work learning cycle** **28**

**B A sample AI-open inquiry cycle** **28**

**C A sample flaw-detection assessment** **29**

**D A responsibility and contribution statement** **29**

# Part I

# The apprenticeship under pressure

## 1 Purpose and premise

What should graduate education in mathematics accomplish when AI can contribute substantially to mathematical research? AI can assist with reading, computation, conjecture formulation, proof construction, and exposition. These capabilities raise questions that go beyond whether students should be permitted to use particular tools. They require us to reconsider how graduate study develops mathematical understanding, creativity, judgment, and independence—and how an apprenticeship should be designed when producing a result, understanding it, and learning to initiate and direct research need not develop together.

The discussion begins from a deliberately moderate premise:

> AI can increasingly produce plausible, technically sophisticated, research-level mathematical material faster than a developing student can understand and check it.

This premise does not require the claim that present systems reliably solve arbitrary research problems. AI-generated proofs may contain hidden assumptions, fabricated references, irrelevant detours, subtle gaps, or fatal errors. That unreliability is part of the educational problem rather than an objection to it. Once a system can generate material whose apparent sophistication exceeds a student's capacity to evaluate it, the traditional connection between performance and formation is already under pressure.

The same capabilities create educational opportunities. Responsive explanation, comparisons among approaches, computational exploration, and assistance with unfamiliar prerequisites may help a student become a more capable mathematician. These benefits are not automatic, any more than intellectual dependence is inevitable. They depend on what the student does, what the system contributes, and how the activity is designed.

The aim is neither to preserve the current PhD unchanged nor to predict a single future. It is to ask how graduate education can cultivate mathematical imagination, understanding, technical power, and independence while making productive use of AI. The proposed practices are designs to evaluate, not claims that one sequence of learning is best for every student, field, or stage.

## 2 The traditional apprenticeship

The relationship between a mathematics PhD student and an advisor often develops through two overlapping stages. Their balance matters because AI may change both the work of formation and the research collaboration it makes possible.

### 2.1 Formation before productivity

Early in the apprenticeship, the student takes courses, learns an area, reads selected papers, follows references, computes, constructs examples, reconstructs proofs, and discusses difficulties with an advisor. The advisor identifies important gaps, chooses worthwhile problems, and helps the student recognize what a successful argument would require.

An assigned problem can be educationally successful even when it is not solved. An attempt may reveal that an estimate fails at the boundary of its claimed parameter range, that an analogy is superficial, or a lemma has indispensable hypotheses. Failed proofs, partial arguments, and well-formed questions can be evidence of developing understanding and research skill. Few publishable results need not imply little progress.

Difficulty and time spent stuck are not educational achievements in themselves. Trying an approach and discovering why it fails can build mathematical capacity; remaining confused because of poor guidance, inaccessible explanations, or an unrecognized prerequisite gap may not. A better apprenticeship should preserve opportunities for the former while providing help with the latter. Human understanding requires intellectual activity and consolidation, but the familiar duration and sequence of that activity are not beyond improvement.

### 2.2 From student to contributor

As students gain command of an area, they read independently, recognize proof strategies, and make sustained contributions. They may identify a missing case, introduce a technique, formulate a better question, or open a direction the advisor had not anticipated. The relationship can become a collaboration and eventually support an independent research program.

This creates a traditional apprenticeship bargain: an early investment in formation can lead to later research collaboration, intellectual continuity, and a future colleague. The relationship is not merely transactional. Advising serves an educational mission even when a student does not advance the advisor's own project, and students contribute through teaching, new perspectives, and the life of the department.

AI may alter this balance in more than one direction. It may reduce an advisor's reliance on a student's routine research contributions, but it may also help a developing student engage sooner in substantive collaboration. Neither the research benefits nor the mentoring costs should be assumed in advance.

## 3 The formation gap

In the traditional apprenticeship, mathematical formation and research production were imperfectly but substantially coupled. Working toward a result required students to recognize gaps, read and experiment, attempt proofs, receive criticism, and revise their ideas. The route to a result was also a route toward the capacity to understand and extend it.

AI makes a shorter route to a mathematical artifact possible: a student can give an AI system a research question and receive a candidate solution or polished draft before undertaking comparable intellectual work. This creates a *formation gap*: the distance between the sophistication of the

available mathematical output and the student's capacity to understand, evaluate, and develop mathematics, including initiating and directing an investigation.

Consider a student who asks AI for help with a research problem and receives a useful approach, a key lemma, and its proof. The student studies the argument carefully, fills in details, and comes to understand it genuinely. When meeting with the advisor, the student explains the proof convincingly and answers the advisor's questions. The project has advanced, and real learning has occurred. Yet the student may not have practiced formulating the lemma, choosing among uncertain approaches, recognizing why a path is failing, or deciding what to try next. If this pattern recurs, those capacities may receive little exercise while progress and convincing explanations suggest to the advisor that the apprenticeship is developing them.

The concern is not that understanding supplied mathematics has no formative value. Studying a proof can develop technique, imagination, and insight, and may itself require substantial creative effort. The concern is that these gains do not automatically establish the student's capacity to conduct an investigation before its direction or answer has been supplied. *The formation gap can persist after the understanding gap has closed.* A good explanation is evidence of understanding, not sufficient evidence of the full range of research capacities.

Presenting generated ideas as one's own adds an integrity problem, but removing that misrepresentation does not remove the educational problem. The student may disclose every substantive contribution, including assistance from AI, the proof may be correct, and the explanation may be excellent, while opportunities to formulate, search, choose, and revise remain repeatedly bypassed. This is a risk to investigate, not a diagnosis of incapacity from AI use alone.

Two judgments must therefore remain distinct: *what the work contributes to mathematics*, and *what the student's participation reveals about their formation.* An important, reliably established theorem does not lose its mathematical value because its proof was produced quickly or with AI. Its value, however, does not automatically certify the maturity of the student presenting it. Conversely, an unsuccessful attempt can develop substantial mathematical capacity without yielding a publishable result.

The appropriate response is not to restore difficulty wherever AI removes it or to require rediscovery of every supplied idea. AI can reorganize the formative process: a student might compare approaches, explore a richer class of examples, investigate a connection, or start from a known answer and seek a new explanation or extension. These activities become occasions for research formation when the student does substantive mathematical work and learns to make and revise consequential choices.

The educational question is therefore not only whether the student understands the result, but where reasoning, experimentation, choice, and invention occur across the apprenticeship. Their distribution among student, advisor, and AI should be deliberately designed for development, rather than left to whichever workflow produces answers most quickly.

## 4 The trust gap

Suppose a student encounters a difficult passage in a reputable paper selected by an advisor. Although the paper may contain mathematical errors—refereeing is not infallible—the student ordinarily has a reasonable working presumption that the passage is mathematically sound. If the student does not initially understand it, sustained effort is likely to be worthwhile. The presumption

of soundness encourages the student to treat confusion as something to work through, thereby turning it into productive mathematical effort.

When the passage is generated by AI, the student may face at least four possibilities:

1. the argument is correct but exceeds the student's present understanding;
2. the argument is basically sound but contains a repairable gap;
3. the argument is false and should be rejected or substantially rebuilt;
4. the argument is formally sophisticated but irrelevant, vacuous, or mathematically uninformative.

The student may lack exactly the knowledge needed to tell these cases apart. Continuing to study a false argument can waste time and create misconceptions; abandoning a correct but difficult argument can prevent learning. Asking AI for clarification may produce another confident explanation rather than independent evidence.

Mathematical trust was never automatic. It was produced by identifiable authorship, editorial selection, refereeing, seminars, citation, correction, independent checking, and the gradual evaluation of published work by a community. These practices did not guarantee correctness, but they gave readers reasons for provisional trust and routes for challenging a claim.

AI-generated mathematics can reach a student before it has passed through comparable checks. Existing vetting practices were designed for public mathematical claims, not for the volume of individualized, real-time output produced during an AI interaction. The student may therefore lack a reliable initial basis for deciding whether the difficulty lies in their understanding or in the argument. The distinction is between evidence and insufficient evidence, not simply between human and machine authorship: a carefully checked AI-assisted proof may deserve more trust than an unchecked human argument.

A confidence estimate from another AI system may help if it has been calibrated against known examples containing relevant kinds of errors. A confidence label alone, however, cannot create warranted trust. A sound learning environment combines reliable references; provenance, meaning a record of where a claim came from and how it was transformed; independent derivations; tests of special cases; counterexamples; symbolic or numerical checks; formal verification where available; peer criticism; and expert review. Students must also calibrate their own confidence, distinguishing between not having found an error and having strong evidence that an argument is sound. The construction and evaluation of mathematical trust must become an explicit part of graduate training.

The kinds of assurance must also be distinguished. Agreement among AI models need not be independent confirmation; examples and numerical tests can expose errors without proving a general claim. A formal certificate establishes a particular formal statement relative to its assumptions and trusted checking system, while the correspondence between that statement and the intended mathematics still requires scrutiny. Students need to learn what each check establishes, not merely accumulate checks.

# 5 The challenges for students, advisors, and graduate programs

The formation and trust gaps create connected challenges at three levels. Students must turn powerful assistance into their own mathematical development; advisors must design and assess a formative apprenticeship; programs must sustain that apprenticeship and justify what their degrees certify. The difficulties are not confined to academic integrity. They arise even when AI use is fully disclosed and the resulting mathematics is correct.

## 5.1 For the student

The student's challenge is to benefit from increasingly capable assistance without becoming dependent on it for the direction and substance of every mathematical activity. An explanation may feel clear while it is being read, yet leave the student unable to use the idea in a different setting. More subtly, the student may understand it deeply while repeatedly leaving the formulation of questions and the choice of approaches to AI. The challenge is to develop both command of supplied mathematics and the ability to begin and sustain inquiry when a promising route has not yet been given.

The student also faces decisions about when to seek help, which suggestions to pursue, and how much of an unfamiliar argument to investigate. The trust gap complicates those decisions: difficulty may reflect a missing prerequisite, a subtle mathematical idea, or a false generated claim. At the same time, assistance can make unfamiliar areas and ambitious projects more accessible. The challenge is to use that expanded opportunity to develop technical skill, imagination, judgment, and increasing ownership of a research direction, rather than merely the ability to obtain a finished answer.

## 5.2 For the advisor

AI changes the evidence available to the advisor, the design of student work, and potentially the balance between research collaboration and educational investment. Four challenges follow.

First, *progress and genuine understanding are incomplete evidence of research formation.* A student may explain an AI-supplied proof thoroughly while gaining little experience in finding a direction before the answer is available. The advisor must distinguish learning to understand an argument from learning to formulate questions, choose approaches, and revise a plan under uncertainty. These capacities overlap, but evidence of one cannot simply stand for all. The challenge is to observe their development without turning every meeting into an examination or an investigation of misconduct.

Second, *the division of intellectual work must be designed for learning.* AI may supply an immediate candidate solution to a problem selected to develop research initiative. Reconstruction and explanation remain valuable, but may not exercise the choices the assignment was intended to develop. The advisor must decide when the student should attempt a formulation before receiving suggestions, when AI should extend or criticize the student's approach, and when it is better to begin from an established answer and investigate its explanation, consequences, or possible extensions. This last possibility is *answer-first inquiry*, developed in Section 9. Continually finding problems beyond AI's reach is neither stable nor necessary. What matters is that students repeatedly practice initiating and directing inquiry, with and without assistance, rather than only understanding work whose direction has already been chosen for them.

Third, *the material requiring expert attention may expand beyond the time available.* A student may bring several generated approaches, lengthy proofs, or techniques unfamiliar to both student and advisor. Assessing their reliability and educational value can require substantial work. The advisor must decide what deserves close review, what the student should investigate first, and when another specialist or verification tool is needed. Without such decisions, advising risks becoming an unbounded review service. AI may also reduce routine explanatory work or assist with checking; the net effect on workload cannot simply be assumed.

Fourth, *the practical incentives supporting intensive mentoring may change.* If AI can perform some tasks previously undertaken by junior collaborators, an advisor may become less dependent on those contributions while still needing to invest heavily in a student's formation. One practical incentive for accepting that investment could weaken. Conversely, AI may enable students to contribute more substantially and sooner, making collaboration more valuable. The challenge is to sustain serious mentoring without assuming either outcome. The educational value of advising does not depend on a student's usefulness to the advisor's own research, but workload and reward structures can still affect whether faculty have the capacity and incentive to undertake it.

Assigning students to check AI-generated output may appear to address both the review burden and the need for useful research contributions. Verification can be intellectually substantive and formative. If routine checking becomes the student's primary role, however, the apprenticeship may develop neither question formulation nor ownership of a research direction. This is why the response cannot consist simply of making students quality-control workers for an advisor's AI-assisted research.

### 5.3 For the program

The program faces a challenge of both certification and institutional capacity. If similar-looking dissertations can be produced through very different levels of student understanding and participation, the artifact alone provides an inadequate basis for certifying mathematical maturity. The program must distinguish a work's contribution to mathematics from the evidence it provides of the student's capacity for independent research. It must also decide what students should understand internally, what they may do with assistance, and what constitutes justified reliance on others.

Those standards must be achievable under the conditions the institution actually provides. More demanding assessment or mentoring cannot simply be added to existing workloads without considering time, expertise, access, and cost. Unequal resources or advising support can make the same formal requirements substantially different in practice. The program's challenge is therefore to make the degree credible while supporting a fair and sustainable route to earning it, with honest expectations about duration and professional opportunities.

The later sections develop responses to these challenges: the redesign of advising in Section 13, evidence of formation in Section 14, and institutional recognition of mentoring in Section 19.

# Part II

# Why human mathematical capacity still matters

## 6 The case for continued formation

If AI can generate increasingly sophisticated mathematical results, proofs, and papers, why should universities continue to invest years of faculty time and institutional resources in developing human mathematicians? Three complementary answers are needed. The first concerns inquiry: trained mathematicians can contribute creatively to discovery, judgment, and synthesis within an AI-assisted research environment. The second is institutional and normative: universities and mathematical communities must preserve the human capacity to understand, teach, evaluate, and take responsibility for mathematical knowledge. The third concerns continuity and resilience: expertise takes years and several generations to develop and transmit, so maintaining it preserves society's ability to respond to an uncertain technological future.

### 6.1 Discovery, judgment, and synthesis

Deep mathematical training supports invention as well as evaluation. It helps researchers formulate questions, recognize structure, devise representations, construct examples, and develop methods. In an AI-assisted environment, these activities may be pursued with a wider range of computational experiments and candidate approaches. The ambition is to prepare mathematicians to use that expanded range intelligently and creatively.

Evaluation and synthesis also remain indispensable functions: someone or some system must distinguish a significant connection from a superficial analogy, assess explanatory power, and relate results to a coherent understanding of a field. If AI-assisted production of candidate proofs and results expands faster than reliable checking and incorporation into mathematical knowledge, these functions become bottlenecks. It does not follow that their growing importance guarantees a growing demand for human labor.

Graduate education should allow for the possibility that AI will contribute increasingly to judgment, question selection, explanation, and synthesis as well as proof construction. The extent and timing of such developments remain uncertain. The argument for graduate formation should therefore not rest on a permanent division in which machines produce and humans judge. It rests in part on the value of people who can participate substantively in mathematical inquiry: form their own questions, work with powerful assistance, assess competing reasons, and develop or redirect a project.

The personal value of understanding and discovery matters too. Mathematics is a human intellectual practice, not only a supply of useful results. Educating people to participate deeply in it is a legitimate university purpose. This does not by itself settle the appropriate scale or cost of graduate programs, but it is a reason that cannot be reduced to a forecast of which tasks humans will perform more cheaply.

### 6.2 Institutional stewardship and intellectual agency

Human mathematical capacity also sustains functions that cannot be measured by paper production alone. Mathematicians referee results, teach the next generation, and maintain standards concerning rigor, proof, evidence, attribution, and the responsible communication of claims. They evaluate the mathematical reliability of AI systems, computational methods, software, and models; serve on scientific, professional, and public panels; and detect when confidently presented errors begin to spread through a research or professional community.

When mathematical claims influence engineering, science, finance, security, medicine, or public decisions, qualified people must be able to determine what has been assumed, what has been established, what remains uncertain, and what consequences may follow if the mathematics is wrong. A community that can request mathematical output but cannot understand, criticize, or independently evaluate it has lost an important form of intellectual agency.

Universities therefore have a stewardship responsibility: important mathematical knowledge must remain open to human understanding, teaching, criticism, and accountability. Stewardship does not require humans to execute every calculation or construct every proof without technological assistance. It requires institutions to retain sufficient expertise to understand consequential mathematical work, evaluate the systems that produce it, challenge unreliable conclusions, and take responsibility for decisions based on those conclusions.

### 6.3 Continuity, resilience, and option value

Mathematical expertise develops through years of study, research, teaching, and collaboration. Once the communities that transmit it have substantially weakened, preserved books and databases do not by themselves restore that capacity. This gives continued formation an *option value*: it maintains the ability to respond as future needs become clearer.

That capacity may matter when systems reveal unexpected limitations, access changes, new scientific problems arise, or consequential applications require independent assessment. It also preserves the ability to develop forms of human–AI collaboration that cannot yet be anticipated. This is not a prediction of a permanent division of labor. It is a commitment to the knowledge and institutional flexibility required to adapt, alongside an obligation to examine the costs and outcomes of that commitment.

## 7 A new educational objective

The central purpose of a mathematics PhD should be to develop the capacity for *original, independent, and responsible mathematical inquiry.* New results remain important, but neither artifact production nor error detection alone defines the objective. A mature mathematician can ask questions, develop ideas, bring a project to fruition, explain its value, and establish a warranted basis for its claims.

Four connected capacities give this objective practical content:

1. **Competence.** The student has substantive mathematical knowledge and technical ability. They can formulate precise statements, construct and analyze arguments, compute, build informative examples and counterexamples, and use definitions and methods in unfamiliar

settings. Competence is productive as well as reconstructive: it supports the invention and development of mathematics.

2. **Judgment.** The student distinguishes conjectures, heuristic arguments that suggest a conclusion without proving it, and rigorously established claims; evaluates assumptions and evidence; compares approaches; and assesses novelty, significance, explanatory power, and usefulness. They can give reasons for pursuing one question rather than another, while recognizing uncertainty and the possibility of revising that choice.

3. **Independence.** The student identifies what they need to learn, formulates worthwhile questions, designs a research strategy, and selects among literature, collaborators, computation, AI, and formal tools. They can initiate, develop, and redirect inquiry rather than merely execute a supplied plan. Independence is compatible with substantial assistance and collaboration.

4. **Responsibility.** The student identifies substantive contributions, explains why central claims should be accepted, describes checks and limitations, and responds to criticism and correction. They distinguish their contributions from those of advisors, collaborators, sources, and AI systems, without treating attribution as a substitute for mathematical understanding.

These capacities include *epistemic responsibility*: an obligation to give a defensible account of what the work establishes and why. Responsibility is necessary for inquiry, but it is not its whole purpose. The formation of a mathematician should also develop curiosity, imagination, technical initiative, and the ability to turn an interesting question into a sustained investigation.

Responsibility does not require a student to reproduce every dependency personally. Modern mathematical work already relies on established results, collaborators, and computational systems. A student should have detailed command of their substantive contributions and the project's central mathematical structure; identify components on which they rely; explain the evidence that warrants that reliance; and know when additional expertise or checking is required. The expected depth varies with the component's importance, novelty, and risk.

This is a demanding but distributed standard. A dependency map or a verified component can support responsible reliance, but neither can replace personal mathematical command. A dissertation committee should be able to distinguish a student who understands and develops a project from one who merely coordinates tools and repeats their conclusions.

# Part III

# A proposed approach to training and advising

## 8 Designing learning with powerful tools

The instructor or advisor, together with the student as their independence develops, should identify the capacity an activity is intended to develop and choose forms of assistance that serve that purpose. An advisor may reasonably prefer a modest problem that develops a student's reasoning to an impressive result the student does not understand. Equally, an ambitious AI-assisted investigation may teach more than a long sequence of exercises whose main difficulty is avoidable technical labor.

Neither a general prohibition of AI nor its use without an educational plan is adequate. Programs need several learning designs: independent attempts followed by hints; study of a complete argument followed by reconstruction and variation; collaborative exploration; and AI-open research with subsequent explanation and evaluation. No single order is appropriate for every topic, student, or stage.

Purposeful independent work remains important. Students need opportunities to generate ideas, experience uncertainty, diagnose obstacles, and decide what to try without immediately receiving a proposed next step. These opportunities should be bounded and responsive. Indefinitely withholding explanations, hints, or feedback from an advisor, peers, or AI can waste time, reinforce misconceptions, or make success depend too heavily on prior preparation.

An activity can specify an initial attempt, the hints or criticism available afterward, when a fuller solution may be consulted, and what reconstruction, adaptation, or extension the student must then undertake. Another activity may begin with a reliable solution and ask the student to explain or transform it. Across activities, however, reconstruction should not become the only kind of intellectual work. Students also need occasions to choose a direction while its prospects remain uncertain. The objective is practice in mathematical decision-making, not a quota of failed attempts or time spent struggling. The test is what the student learns and can subsequently do, not how closely the sequence resembles an inherited apprenticeship.

## 9 Knowing an answer as a beginning

A reliably established answer can initiate substantial mathematical inquiry. Students already learn by studying known theorems; AI-assisted research may create further opportunities to begin from a result and ask why it is true, how far it extends, or what else follows. Knowing an answer can orient these investigations without making them routine or guaranteeing that a simple proof exists.

Consider a student studying a theorem with a technically demanding but independently checked proof. The educational project need not be to rediscover that proof before seeing it. Instead, the student and advisor can ask what mechanism drives the theorem. The student uses AI to propose alternative formulations, candidate weakenings of the hypotheses, and families of examples. These proposals remain conjectural until checked. Some may be rejected by counterexamples; others may lead to a valid extension, a cleaner argument in a special case, or a connection to a different method.

The student then develops one direction, explains why others failed, compares the resulting argument with the starting proof, and identifies what has been learned. Assessment concerns mathematical choices, technical work, explanation, and application to new problems, not ownership of the first answer. The activity may produce new mathematics, or it may be valuable formation without a publishable contribution. Those outcomes should be identified honestly.

The same design can begin with an AI-produced proof, but only after there is sufficient evidence to treat its conclusion as established. A plausible announcement does not give the student a known answer. If the starting proof is unverified, the project is partly an investigation of whether the result is true, and its learning goals should say so.

Answer-first inquiry complements rather than displaces open exploration. Students also need experience choosing questions whose answers are unknown, deciding whether to seek a proof or a counterexample, and abandoning an unproductive formulation. Graduate formation should include both the freedom created by a known result and the uncertainty inherent in research.

## 10 Verification, independence, and community

Verification belongs throughout training. Early activities can test definitions against examples or compare a generated explanation with a textbook. Later activities can analyze limiting cases, construct counterexamples, audit dependencies, or employ symbolic and formal methods. These are not merely inspections of completed work: they can reveal why a result is true, how far it extends, and what questions follow.

The speed of obtaining material must be distinguished from the pace of consolidating it. A summary of several papers is not the same as command of their ideas. Students need opportunities to reconstruct arguments, connect concepts, teach, and apply what they have learned. Retention, application to unfamiliar problems, and increasing initiative are more informative than the quantity of material encountered.

Planning should itself become part of the student's education. Early on, the advisor explains why one paper precedes another or a promising direction is postponed. AI can suggest prerequisites and alternative routes, but suggestions must be discussed and tested. Control should move from advisor design through co-design to student design; meaningful student control may include extensive AI assistance.

Independence does not mean isolation. Seminars, reading groups, teaching, and contact with several mathematicians expose students to different standards and styles. They prevent the output of one system or preferences of one advisor from becoming the entire intellectual environment. The aim is a student who can direct their development while participating in a technological and human community.

## 11 Roles and conditions of AI use

The same system can support or undermine learning depending on the role it is assigned. Four roles are especially useful to distinguish:

1. **AI as tutor.** It explains definitions, changes the level of exposition, produces examples, asks diagnostic questions, and offers hints. The objective is the student's understanding rather than the completion of an artifact.

2. **AI as critic.** It examines work already attempted by the student, points to possible gaps, requests justification, proposes tests, and compares approaches. It may identify that a step is unsupported without supplying the repair.

3. **AI as collaborator.** It proposes strategies, performs calculations, searches for related ideas, drafts proofs, and helps develop a project. The student must track the division of labor and retain responsibility for the mathematics.

4. **AI as object of examination.** It produces proofs, definitions, explanations, or plans that the student must audit, repair, compare, or reject. The possibility of error is educationally central.

These roles may be combined, but they should not be confused. A student does not learn proof construction merely by asking a collaborator for a proof, and a tutor should not quietly become the unacknowledged author of assessed work.

These roles can be explored under different conditions of access to AI. A *role* describes what AI does; an *access condition* describes the assistance available during a particular activity or interval. The following are possible educational designs, not a required classification of every activity, a sequence all students must follow, or a prescription for particular AI systems. They may be useful in ordinary study and research as well as assessment:

| **Condition** | Possible arrangement | Intended educational purpose |
|---|---|---|
| **AI-off** | A bounded interval without generative AI assistance; other available resources are agreed separately. | Develop and demonstrate internal competence, persistence, and live reasoning. |
| **AI-limited** | Selected hints, questions, criticism, or examples, intended to support the student without supplying a complete solution. | Support the student's reasoning while providing responsive guidance. |
| **AI-open** | Broad use of AI for exploration, generation, calculation, checking, and drafting, with appropriate disclosure. | Develop responsible collaboration with AI and pursue authentic or ambitious research. |

Which arrangements are useful depends on the mathematical objective, the student's development, and the setting. An advisor and student might agree on a bounded period of independent exploration, selected assistance while the student develops an approach, or an investigation that begins from a complete argument. These are possibilities to adapt, not a fixed sequence or a new system of surveillance.

Their practical limits matter. An agreement about private study is not proof that a restriction was observed, and asking a model to give hints does not guarantee that it will withhold a solution. Where restricted assistance is important to an assessment, the program needs a practicable arrangement and appropriate accommodations; it should not claim evidence that the arrangement cannot provide. The aim is to support learning and obtain proportionate evidence of development, not to make

every activity fit a category. Broad AI use still requires mathematical responsibility and attention to privacy and equitable access.

Section 14 discusses evidence of formation, and Appendices A and B offer illustrative learning cycles. Their usefulness should be evaluated rather than assumed.

# 12 An integrated training model

Graduate formation should not be organized as "learn first, verify later, research last." Several strands should run through the program at increasing levels of complexity and student control.

## 12.1 Knowledge, technique, and mathematical orientation

Students still need substantial internal mathematical knowledge. Background knowledge determines which questions they can formulate, which analogies they notice, which outputs appear suspicious, and which explanations they can integrate. Foundational courses should remain demanding while distinguishing three legitimate educational goals:

- *working fluency*: definitions, examples, core theorems, and techniques that should be available without extensive external assistance;
- *navigational knowledge*: awareness of areas, methods, mathematicians with relevant expertise, and references sufficient to know what to seek and where to seek it;
- *tool-mediated capability*: computations, searches, formalizations, or specialized procedures that may be performed with external systems, but whose assumptions and outputs the student can evaluate.

Fields will draw these boundaries differently. The important point is that they be chosen for educational and mathematical reasons rather than inherited from current tool limitations.

Proof, computation, and example-building remain central. Constructing a proof teaches more than the final sequence of implications; it develops control of definitions, assumptions, and possible approaches. Examples and counterexamples reveal how objects behave and resist plausible overgeneralization. Students should routinely ask which hypotheses are used, what occurs in limiting or degenerate cases, and whether scaling, dimensional, numerical, or other domain-appropriate checks support a claim.

## 12.2 Guided study and research apprenticeship

The advisor introduces a research area through selected papers, examples, and questions. AI can help map prerequisites, offer explanations at different levels, and generate practice material. The advisor remains responsible for the intellectual design and for explaining the choices. The student presents material independently in meetings, answers questions, reconstructs arguments, and identifies what remains uncertain.

When a research problem is introduced, the advisor, student, and AI may decompose it into subproblems. The decomposition should be visible and contestable: the student should understand

why each part matters and gradually learn to propose the decomposition rather than merely follow it. One cycle moves from an independent attempt to bounded assistance, testing, explanation, and selection of the next question. Another starts from an established result and develops a new direction through AI-open exploration. Appendices A and B illustrate these designs; neither should become a compulsory routine.

The advisor adjusts support based on observed learning. A brief attempt made only to satisfy a requirement before requesting a full AI-generated answer does not demonstrate meaningful learning. Conversely, extensive AI use does not by itself indicate a lack of mathematical development. What matters is the student's substantive participation and the capacities that participation develops. As students mature, they should gain greater freedom to choose their tools and methods, together with greater responsibility for the mathematical decisions that follow.

### 12.3 Progression toward research leadership

The training model describes a progression in responsibility rather than a fixed calendar. At entry, students establish foundational fluency and learn the program's standards for AI use, verification, and attribution. As they enter a research area, they read and present key work, construct examples, and participate in designing their learning plans. In guided research they may explore different arrangements for assistance, develop domain-specific forms of verification, and keep concise records of important decisions and contributions.

In the final stage, the student increasingly selects questions, designs the project, chooses the division of labor, and decides what evidence is sufficient. Success should not be defined as producing something that "AI alone could not have produced," an unstable and unmeasurable counterfactual. The relevant standard is that the student's sustained participation adds identifiable intellectual value and that the student can take responsibility for the final work.

Mathematical maturity also includes communication and teaching. Explaining one result to a specialist, a mathematician outside the area, and an advanced undergraduate reveals whether the student sees the underlying structure or only reproduces notation. Teaching and exposition are therefore not secondary presentation skills; they provide evidence of understanding and prepare students to participate in a community of knowledge.

## 13 The advisor in the new model

The advisor helps students develop mathematical imagination, technical command, taste, and research independence. This includes selecting reliable sources, discussing worthwhile problems, making the reasons for a research choice visible, and recognizing when a student's unexpected direction deserves support. The advisor also distinguishes genuine understanding from fluent output and models how to use AI critically and creatively. Establishing that the student genuinely understands an argument is not enough: the advisor must also consider whether the student is developing the ability to initiate and direct mathematical work.

The frequency of research meetings need not change. What needs reconsideration is the intellectual activity between meetings and the evidence discussed within them. Alongside completed arguments, meetings should sometimes address an unfinished question: what does the student think is worth

trying, what alternatives are plausible, and what evidence would justify changing course? Discussing choices before their outcomes are known provides different evidence from a polished retrospective explanation.

For a selected task, student and advisor can identify the capacity to be developed and agree on a suitable division of work. The student might formulate candidate lemmas before asking AI for criticism, compare assisted approaches and choose one to pursue, or start from a supplied proof and design an investigation of an unresolved extension. A brief note of a consequential choice and its reasons can support the discussion; routine progress need not generate a new reporting burden. The next meeting considers both the mathematics and whether the activity exercised the intended capacity.

These practices do not make the advisor a monitor of private study or require withholding help until the student fails. They make AI use part of the deliberate design of the apprenticeship. If existing advising already provides these opportunities and credible evidence of development, it may need little structural change. What cannot be assumed is that project progress plus genuine understanding establishes that all is well.

AI may reduce some routine demands and increase others. Generating practice material or clarifying prerequisites may save time, while reviewing sophisticated generated arguments may require more. The balance should be observed rather than assumed. Whatever the division of labor, the student needs substantive mentoring, professional advocacy, and access to mathematical perspectives beyond one system or one advisor.

An AI-generated study plan may explain why a prerequisite should be studied first, why one paper is relevant, or why a particular problem is a useful next step. Those explanations can support learning, but the advisor and student should still discuss whether they fit the student's needs and research goals. The aim is not to reserve planning for humans, but to develop the student's ability to evaluate, revise, and eventually direct a plan.

Preparing faculty members to advise and teach in this environment should involve actual mathematics: exploring AI-assisted approaches, comparing proofs, designing activities with different kinds of assistance, and discussing how to assess understanding fairly. It should also address attribution, confidential material, unequal preparation, and the distinction between educational observation and surveillance. Learning records should support feedback, not demand continuous monitoring of a student's private work.

The advisor must protect the educational purpose of the relationship. Verification is part of mathematical inquiry and may itself be creative research, but assigning students routine checking of an advisor's generated output is not an adequate apprenticeship. Their work should progressively include question formulation, method development, and ownership of a research direction.

# Part IV

# Assessment and program redesign

## 14 Assessing capacities rather than artifacts alone

Research results remain valuable. They can show ambition, persistence, and the ability to bring a project to completion. They can no longer serve as the primary proxy for student development. Programs need direct evidence of competence, judgment, independence, and responsibility.

| **Capacity** | Illustrative evidence | What the evidence helps reveal |
|---|---|---|
| **Competence** | Proof construction and reconstruction, examples, computation, development of a new approach, application to unfamiliar problems | Whether knowledge and technique support productive reasoning |
| **Judgment** | Flaw detection, comparison of approaches, confidence with reasons, evaluation of significance and naturalness | Whether the student can distinguish plausibility, correctness, and mathematical value |
| **Independence** | Student-formulated questions, research plans, creative use of assistance, choice and revision of methods | Whether the student can initiate and develop inquiry |
| **Responsibility** | Contribution statement, checks of central claims, account of uncertainty, oral defense of decisions | Whether the student can understand and stand behind work presented under their name |

No single instrument can measure all four capacities. Assessment should combine work produced over time with direct demonstrations under clearly described conditions. The possibilities in Section 11 can serve different purposes where they are practical: AI-off work may show foundational fluency and live reasoning; AI-limited work may show how the student uses guidance; AI-open work may show responsible collaboration in an authentic research environment. No single activity must use all three arrangements.

Take-home work remains useful for assessing exposition, synthesis, revision, and tool use, but it becomes weak evidence of unaided competence when considered alone. A short oral follow-up, whether part of an examination or an ordinary meeting with the advisor, can ask the student to explain a key step, adapt the solution to a changed hypothesis, identify where assistance entered, or reconstruct the argument without the submitted text. Oral examination should not become the sole safeguard: quick verbal performance can reflect language background, disability, anxiety, and style as well as understanding. Observations across regular advising meetings and seminars, written work, and sustained projects provide complementary evidence. Assessment should include appropriate accommodations rather than depend on one style of performance.

Even an excellent explanation or reconstruction cannot by itself establish research initiative. Evidence should also include work on questions before a route has been supplied: a proposed formulation, reasons for trying an approach, a revision prompted by an obstacle, or the design of a next investigation. Some of this work can be AI-open; the issue is the student's substantive participation in making decisions, not the absence of help. Discussing what the student intends to try before the outcome is known, and observing how they approach new tasks over time, provides evidence that a later account of a completed project cannot supply by itself. No decision log or single successful demonstration establishes independence on its own.

## 15 Exercises in judgment and creative inquiry

Several assessment practices are especially suited to an environment filled with plausible AI-generated mathematics. They can be used formatively before they carry high-stakes consequences.

In a *flaw-detection exercise*, an instructor begins with correct arguments and introduces controlled errors: a dropped hypothesis, reversed quantifiers, an unjustified extension to a boundary case, an incorrect dependency of a constant, an unjustified exchange of limits, an unhandled case, or a vacuous conclusion. Students locate and classify the issue, explain why it matters, state their confidence, and repair it if possible. Correct arguments must be included as negative controls; otherwise students are rewarded for inventing objections to every proof. Results should distinguish detection from false positives and should be examined by error category.

Confidence itself can be trained. For selected judgments, students report a coarse confidence level together with the evidence behind it. Across many tasks, the program can help students see whether they are systematically overconfident, or whether they identify genuine problems but distrust correct arguments as well. The objective is not to attach decorative percentages to statements, but to develop calibrated self-knowledge.

Other useful exercises ask students to remove a hypothesis and construct a counterexample, analyze a boundary case, compare several correct approaches, rank research questions by significance and defend the criteria, or explain the same result at several levels. Such activities reveal forms of understanding and judgment that a polished final artifact may conceal.

Creative tasks should be equally visible. Students can formulate a conjecture from examples, design an informative computational experiment, propose two approaches to a question, or develop a connection suggested by AI. Assessment should consider the quality of the question, mathematical reasoning, use of evidence, and productive revision, without rewarding confident novelty claims over careful investigation.

## 16 Milestones, examinations, and the dissertation

Qualifying examinations should state what they certify. AI-off components can test foundational fluency, AI-open components can test synthesis and productive tool use, and oral or written follow-ups can test understanding and adaptation. No single mode should carry the entire burden of establishing mathematical maturity.

Candidacy should demonstrate the ability to enter a field, understand and examine a substantial argument, formulate a worthwhile question, and design a plausible investigation. A proposal should explain the role of AI and other resources, what uncertainties remain, and how the student will respond to evidence that the original direction is unpromising.

### 16.1 Originality and the student's contribution

The dissertation should make a substantial contribution to mathematics and demonstrate the student's capacity for independent research. These are related but distinct requirements. An important AI-assisted result does not automatically establish the student's maturity; excellent understanding of existing work does not by itself constitute an original research contribution.

Originality need not be confined to obtaining the first proof of a new theorem. A new method, a significant generalization, an illuminating alternative proof, a conceptual connection, or a mathematically substantive formalization can advance a field. These are possibilities for evaluation, not automatic qualifying categories. Routine formalization, competent exposition, or a useful compilation may be valuable without meeting the standard for a doctoral research contribution.

The committee should identify what is new, why it matters, and how the student contributed intellectually to its development. Contributions may include problem formulation, mathematical design, technical arguments, interpretation, or a decisive revision of an approach. They must be substantial and demonstrable across the project, not merely a claim to having initiated an AI interaction or assembled its outputs.

The standard is not that the student has done something AI could never do. It is that the work advances mathematics and the student's sustained participation demonstrates original, independent, and responsible inquiry. As elsewhere in collaborative research, credit for a shared achievement and evidence of an individual's formation require careful, explicit judgment.

### 16.2 Evidence, defense, and proportionate records

The dissertation should be accompanied by a concise responsibility and contribution statement addressing main claims and important decisions rather than every sentence. It identifies substantive human and AI contributions, relevant sources, checks, dependencies, unresolved limitations, and enough information to examine important computational or formal components.

The defense tests whether the student can explain, modify, criticize, and contextualize the work. A committee might change a hypothesis, request a special case, ask how a new direction was chosen, or examine a component on which the project relies. It should also test creative understanding: what further question does the work make possible, and what would a serious attempt to answer it involve? These demonstrations should be considered alongside evidence of initiative and changing responsibility during the research, not treated as a substitute for it.

Records should remain proportionate. Useful details may include the name and release of an AI model, the versions of computational software or proof assistants used, relevant settings, and selected interactions that materially shaped the work. An archive of every query is neither necessary nor sufficient. Repeating a query may not reproduce the same response, and an earlier model may no longer be available. What must remain open to examination is the mathematics itself: the proof or code actually used, its assumptions and dependencies, and the checks that support its conclusions.

Where supporting material is confidential, authorized examiners should have a secure way to review what they need without requiring that the material be made public.

## 17 Curriculum, policy, and community

Core mathematical education should not disappear because information is retrievable. Courses should identify the concepts and techniques students need internally, emphasize connections and the application of learned ideas to unfamiliar problems, and choose forms of AI assistance for specific educational purposes. Breadth may become more important, not less, when students must evaluate connections and generated analogies across areas.

Verification and judgment should form a thread through the program rather than a single new course. A first-year class can use elementary flaws and counterexamples; an area seminar can examine research-level arguments; an advanced group can referee AI-assisted drafts or use formal and computational checks where relevant. Supervised peer-review seminars can distribute practice economically, but faculty must set standards, audit the discussion, and correct shared errors.

Programs also need task-specific policies. A single rule cannot govern homework, qualifying examinations, reading courses, and open research equally. Policies should state the permitted role of AI, disclosure expectations, data restrictions, responsibility for errors, and consequences of misrepresentation. Faculty should disclose their own substantive AI use when collaborating with students; responsibility cannot be a one-way demand. AI systems can respond to criticism and help revise an argument, but those responses do not discharge the professional accountability of the human authors or committees who endorse it. Institutions must assign responsibility for investigating errors, correcting the record, and protecting affected students.

Equal access to an interface does not guarantee educational equity. AI can provide responsive explanation to students with uneven prior opportunities, but those with weaker preparation may also be least able to detect plausible errors. Paid tools, computing access, disability, language, and variation in advisor engagement can divide a cohort. Programs should provide a supported baseline of AI and computing access, early verification training, multiple forms of assessment, human support for foundational gaps, and routes for challenging automated judgments. Access to AI and computing resources means sufficient usage allowances, time on shared systems, and technical support, not merely an account. Shared university or consortium provision can support ordinary work, with an additional allocation process for projects requiring substantially more computing time, memory, or specialized technical support. Assessment must not silently depend on a student's ability to purchase more powerful assistance.

Finally, programs should resist the apparent efficiency of replacing seminars and human discussion with individualized tutoring. Students need to see experts disagree, watch questions being formulated, learn disciplinary norms, and encounter several styles of reasoning. Cross-area seminars, rotations, reading groups, teaching, and informal conversation remain central to the formation of judgment.

## 18 Duration, scale, and professional preparation

The case for automatically shortening the PhD assumes that its product is research output. If the purpose is mature mathematical inquiry, the binding constraint may be human consolidation

rather than generation speed. Reading, forgetting, returning, teaching, failing, and revising are not obviously compressible because an AI system can draft a proof quickly.

Programs should become more competency-based and less dependent on a uniform chronology. A well-prepared student who demonstrates the required capacities should not be held only to satisfy a customary number of years; a student should not be rushed because AI made a dissertation draft arrive early. Competency-based progression requires funding and clear milestones so that "maturation takes time" does not justify indefinite enrollment.

Program scale is a related but separate question. Preserving a pipeline of trained mathematicians does not determine a particular cohort size, nor does AI provide an automatic argument for reducing it. Departments must consider advising capacity, educational quality, career outcomes, and the full costs of student support, mentoring, teaching provision, and computational infrastructure. Maintaining or expanding a cohort may be justified by the institution's educational and research mission, the need to sustain expertise, and credible opportunities for graduates, provided the necessary support can be funded. Where support is inadequate, the alternatives include additional investment, shared resources, and redesigned advising as well as changes in admissions. Neither preserving existing numbers nor shrinking them substitutes for examining what the program can responsibly provide and what purposes it should serve.

The changing profession need not mean only declining opportunity. As AI produces more mathematical arguments, models, computations, and conjectures, new needs may arise for people who can understand, organize, evaluate, and responsibly apply that output. Alongside existing careers in university research and teaching, industry, and public institutions, mathematically trained graduates may contribute to emerging or expanded forms of work. These include translating AI-generated ideas into reliable scientific, engineering, and computational methods; providing mathematical oversight in consequential applications; supporting interdisciplinary modeling; developing AI-assisted mathematical education; and leading long-term human–AI projects requiring problem selection, verification, synthesis, and interpretation. Some of these activities already belong to mathematical careers; AI may change their scale, content, or institutional setting rather than create entirely new occupations.

The precise shape and scale of these roles cannot yet be known. Programs should not promise them as guaranteed career paths, but neither should they assume that advanced mathematical training will lose its value. Students need honest discussion of risks and possibilities together with experience in rigorous reasoning, computation, formalization where appropriate, reproducible work, technical communication, collaboration, project design, and learning unfamiliar domains. Career preparation should be integrated into authentic projects, teaching, interdisciplinary collaboration, or internships rather than added as a generic workshop at the end of the degree.

# Part V

# Institutional responsibility and implementation

## 19 Recognizing the work of advising

Faculty performance is generally evaluated through research, teaching, and service, yet graduate advising does not fit neatly within any one category. In many departments it is counted as service, treated as an additional educational responsibility on top of the formal teaching load, valued indirectly as evidence of an active research program, or recognized through some combination of these approaches.

These conventions may become inadequate as the work of advising changes. Some activities may become less demanding, others more intensive, and AI-assisted students may contribute to research in new ways. Workload policy should reflect actual mentoring rather than presume either an automatic saving or an inevitable increase. Where the intensive work of developing students' mathematical capacities is necessary but unrecognized, faculty face an incentive to withdraw from advising or provide only nominal supervision. Departments and universities should reconsider how advising is credited in workloads, annual reviews, salary decisions, and promotion.

One possible mechanism is a *graduate mentoring allocation*. A department defines the activities that constitute intensive formation—an individual learning plan, regular substantive meetings, opportunities to develop research initiative, assessment of understanding and mathematical choices, feedback on verification, and an annual development review—and assigns documented workload credit. Accumulated credit may produce a course release, summer support, reduced committee work, or another locally meaningful adjustment. Co-advisors divide rather than duplicate the credit, and promotion review treats the mentoring record as educational contribution.

This mechanism has real costs. A course release requires replacement teaching, fewer offerings, larger classes, or additional budget. The cost should be visible rather than transferred silently to individual faculty. Team advising, shared area seminars, and verification workshops can reduce duplication and expose students to several standards, but they also require coordination. If a department cannot support responsible formation for its current cohort, it should examine additional investment, shared advising, and changes in program design alongside future admission numbers and program promises. Existing commitments to students should be protected; reducing admissions is not the only response to inadequate support.

## 20 A staged departmental implementation

A department need not redesign the entire PhD at once, but staged implementation should organize prompt action rather than postpone it. Departments can begin discussing AI use in advising, clarifying responsibilities, supporting equitable access, and examining evidence of student development while more specific educational practices are tested. The stages below can overlap; they need not become a sequence of lengthy approvals.

A first stage can establish a common language for the four capacities in Section 7 and the four AI roles and possible access conditions in Section 11. Two existing graduate courses might each try a purposeful independent attempt, a flaw-detection exercise, and an AI-assisted creative investigation, choosing practical arrangements suited to their goals. One activity could begin with a reliably established answer and assess the student's ability to explain, vary, or extend it. Selected take-home work can be paired with brief oral follow-up. A departmental working session for advisors and instructors could examine AI-generated arguments from their own fields, compare their educational uses, and discuss how to assess the student's contribution and understanding.

A second stage can adapt promising practices for milestones: a qualifying component that draws on more than one practical arrangement for assistance, a candidacy exercise in auditing and research planning, and a contribution statement attached to advanced projects. The department can establish a supervised student-refereeing seminar or incorporate peer review into an existing seminar. At the same time, it should review tool access, privacy rules, accommodations, advisor training, and the workload credit attached to early-stage formation.

A third stage can reconsider the curriculum and program as a whole in light of what the earlier stages reveal. This may include changes to course requirements, breadth, dissertation expectations, degree duration, cohort size, and professional pathways. The distinction is between acting promptly and committing prematurely. Departments should make reversible improvements now, set explicit dates for reviewing their effects, and revise them as evidence accumulates. Changes with lasting consequences—especially those affecting degree requirements, admission commitments, or student funding—need broader deliberation and protection for current students. The aim is timely action with accountable revision, not an indefinite pilot or a demand for certainty before anything changes.

## 21 Learning from a pilot

The model contains empirical assumptions: that purposeful independent work supports retention and the use of learned ideas in new settings, that verification practice improves judgment, and that guided AI collaboration can develop both mathematical initiative and effective tool use. These assumptions should be examined rather than embedded permanently in policy.

A first pilot should evaluate components, not compare tiny graduating cohorts trained under entirely different regimes. In courses or reading groups, a pilot might compare matched tasks under selected AI-off, AI-limited, or AI-open arrangements where those arrangements can be implemented credibly. Students can encounter the arrangements in different orders to reduce the influence of practice or sequence. The pilot need not include all three. Another comparison could examine an initial attempt versus beginning with an established solution. Tasks should be comparable in difficulty; all students should have access to the learning opportunities and appropriate accommodations.

Evaluation should distinguish three outcomes:

1. **Immediate mathematical work:** correctness, quality of explanation, quality of questions or approaches, useful revisions, and time required.
2. **Internal formation:** reconstruction after a delay and application to unfamiliar problems without the original output, together with formulation and choice of approaches on a new question before a route is supplied.

3. **Assisted research capability:** later work on an unfamiliar problem with AI available, assessing how students choose assistance, test suggestions, develop ideas, and revise direction.

The third outcome matters because a program devoted to human–AI inquiry should not evaluate success exclusively through unaided performance. Conversely, a strong assisted result cannot alone establish what a student has internalized. The two kinds of evidence are complementary.

Judgment tasks should track false positives as well as error detection, with results examined by error category. Creative tasks need explicit criteria and examples of acceptable reasoning, so that evaluation does not merely reward an assessor's preferred style. Where feasible, assessors should review work without knowing its instructional condition, and disagreements should inform refinement of the criteria.

The pilot should also record student burden, advisor and staff time, computational resources, and differences associated with preparation or access needs. A practice that works only with exceptional supervision may need redesign before broad adoption. Research participation and data use require appropriate institutional review and consent procedures; ordinary educational support should not depend on agreeing to additional research data collection. Small samples and changing systems limit what can be concluded, so the conditions and relevant tools should be documented.

Before the pilot, the department should state how findings would affect its plans. If independent attempts improve later application to new problems at reasonable cost, they can be expanded where effective. If AI-open work produces equal or better formation, restrictions should be narrowed. If one design improves assisted investigation while weakening foundational command, the program should investigate how to combine its benefits with targeted practice. If flaw-detection work raises both valid detections and false accusations, calibration and evidentiary standards need attention.

Such a pilot cannot establish the value of an entire PhD or predict careers. It can identify promising practices, expose attractive assumptions that fail, and reveal the resources needed for responsible implementation.

## 22 Conclusion

AI makes it necessary to distinguish more carefully between producing mathematical work, understanding it, and developing the capacity to initiate and direct inquiry. A valuable result and a genuinely understood proof can coexist with insufficient practice in formulating questions and choosing approaches under uncertainty. But assistance can also expand the student's opportunities for exploration, explanation, and discovery. Graduate education should deliberately shape that participation rather than infer complete formation from progress and understanding alone.

The central objective is original, independent, and responsible inquiry. It requires substantive knowledge and technical skill, the imagination to formulate questions and develop approaches, judgment about evidence and significance, and the capacity to collaborate without becoming a passive recipient of conclusions. Mathematical maturity is demonstrated in what a student can initiate, understand, build, explain, revise, and responsibly contribute.

A redesigned apprenticeship should therefore combine purposeful independent work, learning from established answers, creative AI-open projects, and continuing practice in verification. It should distinguish mathematical contribution from evidence of formation, make doctoral originality explicit, and permit justified reliance on others without surrendering personal command. Institutions must provide equitable support, recognize the actual work of advising, and assess whether their educational designs succeed.

The aim is not to preserve difficulty for its own sake or to locate a shrinking set of tasks beyond the reach of AI. It is to educate people who can participate deeply in mathematics as its possibilities expand. The methods should remain open to revision; the ambition is a richer formation that enables mathematicians to ask more, understand more, and contribute more through their own work and their collaboration with powerful tools.

## AI assistance disclosure

This article was developed with assistance from ChatGPT (OpenAI) and Claude (Anthropic). These tools were used to revise text, to test arguments and supply counterarguments, and to help locate and check sources. The author determined the positions advanced here, made the final decisions on all text, and is responsible for the content.

# A A sample protected-work learning cycle

The following cycle combines protected work, AI assistance, verification, and advisor feedback for a student entering a research area. Its timing and difficulty should be adjusted to the student and material; it is one design, not a required schedule.

1. **Orientation.** The student reads the advisor's statement of the activity's objective and writes what they already know, what appears unfamiliar, and why the topic matters to the larger plan.
2. **Protected reading and attempt.** The student reads selected primary or textbook material and reconstructs a central example or proof without AI assistance for a specified interval.
3. **Tutor or critic interaction.** AI asks questions, supplies alternative examples, or gives bounded hints about points the student has identified.
4. **Independent reconstruction.** With the interaction closed, the student rewrites the argument, performs a related computation, or applies the idea to a new problem.
5. **Audit.** The student compares the reconstruction with trusted sources, tests boundary cases, and records what remains uncertain. A short generated argument may be included specifically for criticism.
6. **Advisor or group meeting.** The student explains the mathematics without a generated script and discusses a choice or obstacle, including one unresolved question and what they would try next. The advisor assesses understanding and initiative, then adjusts the plan.
7. **Increasing student control of planning.** The student proposes the next objective and the appropriate access condition. Early in the program the advisor may substantially revise the proposal; later, the student should control it.

# B A sample AI-open inquiry cycle

This cycle complements the protected-work design. It can begin with a trusted theorem or with an open question; the mathematical status of the starting material must be clear.

1. **Choose a direction.** The student states a question and why it matters. If beginning from a known theorem, they identify a possible extension, explanation, alternative proof, or consequence worth investigating.
2. **Explore with assistance.** The student uses AI, literature, computation, or collaborators to develop examples and candidate approaches. Generated claims are recorded as proposals, not established facts.
3. **Make mathematical choices.** Before the outcome is known, the student selects a direction and gives reasons for trying it. They do substantive work, revising the plan as evidence develops. This may include proof construction, counterexample design, a new formulation, or a decisive comparison.
4. **Establish and explain.** The student checks what the investigation supports, identifies dependencies, and separates established conclusions from unresolved questions.
5. **Discuss and extend.** In a meeting, the student explains the main decisions, responds to a changed assumption or new objection, and proposes a next question. The advisor assesses both personal command and effective collaboration.

A concise record of consequential decisions is more useful here than a complete interaction log. A project that yields no new theorem can still show significant formation; any claim to new mathematics must be evaluated separately.

# C A sample flaw-detection assessment

An assessment set contains correct and modified arguments matched by topic and approximate difficulty. For each argument the student submits:

1. a verdict: no identified problem, minor issue, repairable gap, serious problem, or unable to determine;
2. the exact location and category of any issue;
3. an explanation or counterexample;
4. a possible repair, if one is available;
5. a coarse confidence level and the evidence supporting it;
6. the additional information or check that would most reduce uncertainty.

Possible injected categories include dropped or weakened hypotheses, swapped quantifiers, incorrect constant dependencies, invalid boundary cases or scaling, unjustified exchanges of limits or derivatives, unhandled cases, circular dependencies, invalid citation use, vacuous conclusions, and irrelevant arguments that do not establish the stated claim.

The scoring report should separate localization, diagnosis, severity, repair, confidence, and false-positive rate. The instrument should be used formatively before it is used for high-stakes evaluation.

# D A responsibility and contribution statement

For a dissertation chapter, paper, or major project, the student can address the following questions:

1. **Purpose and significance:** What question does the work address, and why is it worth addressing?
2. **Human contributions:** Which central ideas, decisions, proofs, computations, explanations, and revisions came from the student, advisor, or other collaborators?
3. **AI contributions:** Which systems were used, in which roles, and for which substantive components?
4. **Sources and novelty:** How were related results located, and how was the relationship to existing work assessed?
5. **Verification and reliance:** For each main result, what checks were performed, and what remains uncertain? Which dependencies does the student understand in detail, which are relied upon, and what warrants that reliance?
6. **Rejected paths:** What important generated suggestions were rejected or substantially changed, and why?
7. **Limitations:** Which assumptions, boundary cases, dependencies, or open gaps should a reader know about?
8. **Reproducibility and privacy:** What supporting material can be shared, and what cannot be disclosed because of privacy, licensing, confidentiality, or size?